AFTERSHOCK

Perception-driven, safety-gated disaster-response robotics | AgentVerse 2026 | Technical summary

PROBLEM STATEMENT

Earthquake-damaged buildings expose rescuers to unstable debris, secondary collapse and restricted access, like in the 2008 Sichuan earthquake of magnitude 7.9 with 69,196 confirmed deaths, 18,379 missing, and over 374,000 injuries. People we know and love are affected by natural disasters, so we feel the responsibility of helping others facing similar hardships.

WHY A ROBOTIC AGENT?

Rapid survivor localisation is critical, but unstable structures, hidden hazards, and uncertain casualty locations make rescue operations dangerous. AFTERSHOCK simulates a Scout robot mapping hazards and locating survivors, followed by a Responder navigating physics-validated routes to deliver essential supplies. A simulated aftershock tests the robots' ability to adapt to changing conditions.By gathering critical information before human entry, robotic agents aim to **reduce risks to rescuers and support faster, safer rescue decisions**, rather than replace human rescue teams.

KEY FEATURES

Observe: LiDAR/depth mapping, thermal + acoustic + optional vision casualty cues, observed/inferred/unknown map cells.

Decide: confidence-aware exploration, hazard estimates, local A* route planning and physics verification.

Adapt: an aftershock invalidates the old route; the Scout re-scans and the planner re-routes.

Act & explain: simulated Responder delivers supplies, operator approves risky actions, browser shows evidence and reasoning, voice/typed interaction and an auditable handover report.

SYSTEM ARCHITECTURE

MuJoCo scene + Scout sensors **LiDAR / depth / thermal / audio**	**Perception** **Map, casualty confidence, hazard / motion zones**	**Planner + verifier** **A* route, MuJoCo physics, human gate**
Responder simulator **Move / grip / supply placement**	Async topic bus + FastAPI Validated events; WebSocket /ws	Operator console + voice 3D map, approvals, reports, speech

MODEL PLACEMENT, HARDWARE & LATENCY

Model / function	Execution location and hardware	Latency / budget
Local A* + safety gate	Local Python; laptop CPU, 8 GB RAM	Measured <50 ms per planning/gate step
Gemma 4 E2B (offline goals)	Ollama Q4_K_M; 8 GB RAM laptop, no GPU; 4.1 GB resident	Measured 1.8–2.8 s warm; ~19 s cold; 4 s step target
Gemini 4 Argon (reasoning / triage)	Cloud API; local fallback when unavailable	Not measured; not on critical motion path
Gemini 3.8 Flash (vision / report)	Cloud API; vision triggered at most 1/s	End-to-end latency not measured
Gemini 3.8 Live / Flash TTS; 3.5 Live Translate	Cloud API for speech / translation; text fallback	End-to-end latency not measured

CRITICAL FAILURE MODE & SAFETY

Failure mode: an aftershock moves debris after a route has been approved, making a previously safe corridor unsafe.

Mitigation: sensed change revokes the route permit and stops/holds simulated motion; new Scout measurements, hazard exclusion zones and a fresh physics check are required before movement resumes. Unknown/inferred cells cannot be treated as confirmed safe. High-risk actions need explicit human approval; denial or a 60 s timeout leaves the Responder idle. Approval cannot override failed verification. Emergency calls are prepared, not automatically placed.

LIMITS & EVIDENCE

Simulation-based proof of concept: sensor streams, casualties and vital signs are simulated; structural risk is an estimate, not engineering certification. Mapping may miss occluded or elevated debris; routes require robot-footprint and real-world validation. Cloud model latencies have not been benchmarked, and live services may be unavailable. Hardware (Jetson-class Scout/Responder, cameras, LiDAR, watchdog and e-stop) is a future design, not deployed or safety-certified.

TRACK FIT

Shows sensor-driven action and visible replanning after change.

Video: sensing beside action, one unscripted change, one take, up to one minute.